ModelIt!

Student Activity Pack

The Vaping Crisis: What's Really Happening to Your Lungs?

Modeling Chemical Damage to the Respiratory System

Grade 09L1 | NGSS HS-LS1-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think is actually inside vape aerosol — is it just water vapor?

2 Why do you think vaping-related lung injuries have become so common in teens?

3 Draw what you think happens to lung cells when exposed to vaping chemicals.

4 How do you think the body responds to foreign chemicals entering the lungs?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Nicotine Concentration | Chemical Exposure Duration | Lung Cell Health | Inflammation Response

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Occasional Use

Low nicotine, short duration

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Daily Heavy Use

High nicotine, long duration

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Quit After 6 Months

High exposure followed by cessation

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Oxygen Exchange Rate — The efficiency of gas exchange in the alveoli, which decreases as cells are damaged and airways become inflamed

Mucociliary Clearance — The lung's self-cleaning mechanism using mucus and tiny hair-like cilia to sweep out particles and pathogens

Immune Cell Activity — The concentration and behavior of white blood cells in lung tissue, which increases with chronic chemical exposure

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If vaping is "just water vapor," why are teenagers ending up in the hospital? Use evidence from your simulation to support your answers.

1. **How does your model show the difference between short-term and long-term effects of vaping?**

2. **Why does the inflammation response — which is meant to protect you — end up causing more damage?**

3. **What does your model predict about lung recovery if someone quits vaping after one year versus five years?**

4. **How might your model change if you added oxygen exchange rate as a component?**

5. **What are the limitations of modeling complex biological damage with only four components?**

8 STEM Challenge: Design a Vaping Impact Public Health Campaign

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YOUR ENGINEERING MISSION

Create a science-based public health campaign that uses data from your model to show teens the real biological impact of vaping on lung tissue.

THE SCENARIO

The local health department has noticed a 40% increase in teen vaping and wants to launch a campaign that goes beyond 'just say no.' They need scientifically accurate materials that show WHAT actually happens inside the lungs. Your team has been hired to create the campaign using your model data.

Engineering Design Thinking

1. WHAT FORMAT WILL YOUR CAMPAIGN TAKE — SOCIAL MEDIA, POSTER, VIDEO, INTERACTIVE WEBSITE?
2. HOW WILL YOU VISUALIZE THE DAMAGE CASCADE FROM YOUR MODEL IN AN ENGAGING WAY?
3. WHAT DATA FROM YOUR SIMULATIONS WILL YOU USE AS EVIDENCE?
4. HOW WILL YOU ADDRESS THE MISCONCEPTION THAT VAPING IS HARMLESS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Pulmonologists are physicians who specialize in respiratory diseases and lung health. They diagnose and treat conditions from asthma to EVALI (E-cigarette/Vaping-Associated Lung Injury), earning

$300,000–$500,000/year. Respiratory therapists who work directly with patients earn $55,000–$75,000/year.